DIODE

THEORY OF OPERATION

prepared by:

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Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

The Unbiased Diode

The Biased Diode

- Reverse Bias and Forward Bias

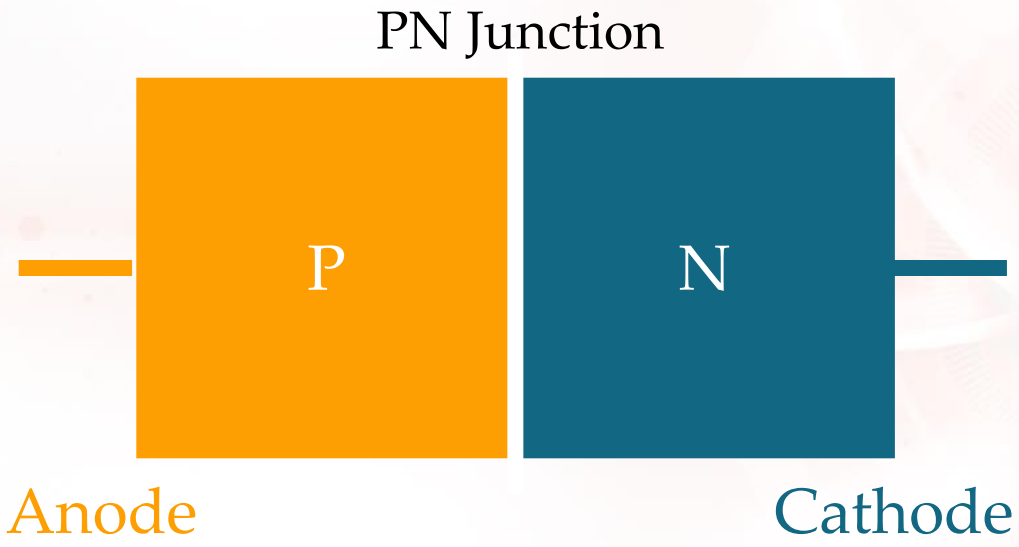
Diode Parameters

Rectifier Circuits

- Half-Wave and Full-Wave

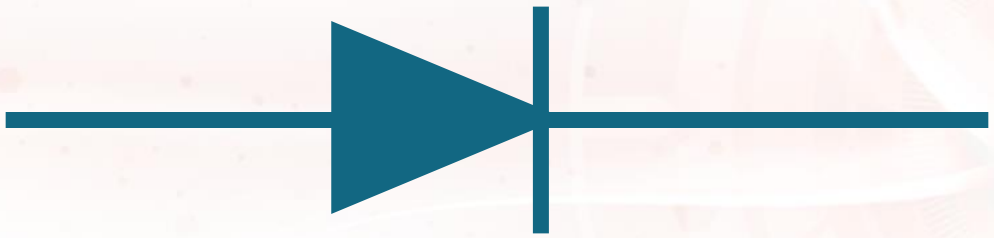
THE UNBIASED DIODE

Basic Construction



The **boarder** between the P-type and N-type material

Schematic Symbol



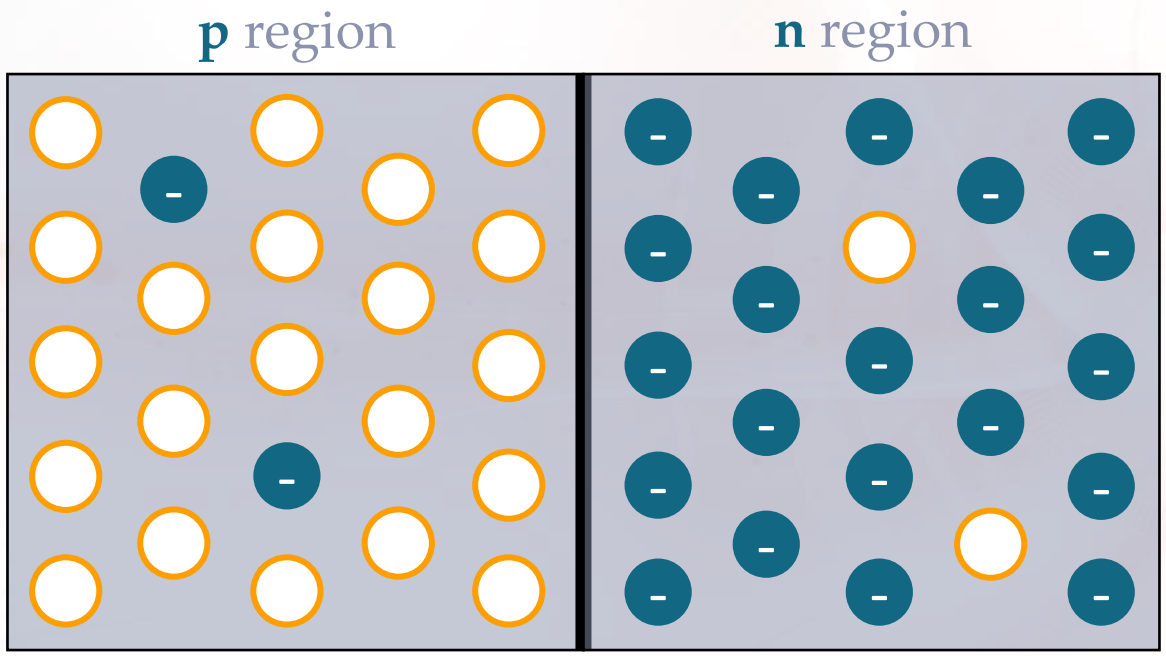
Anode

Cathode

"**Di**" stands for two

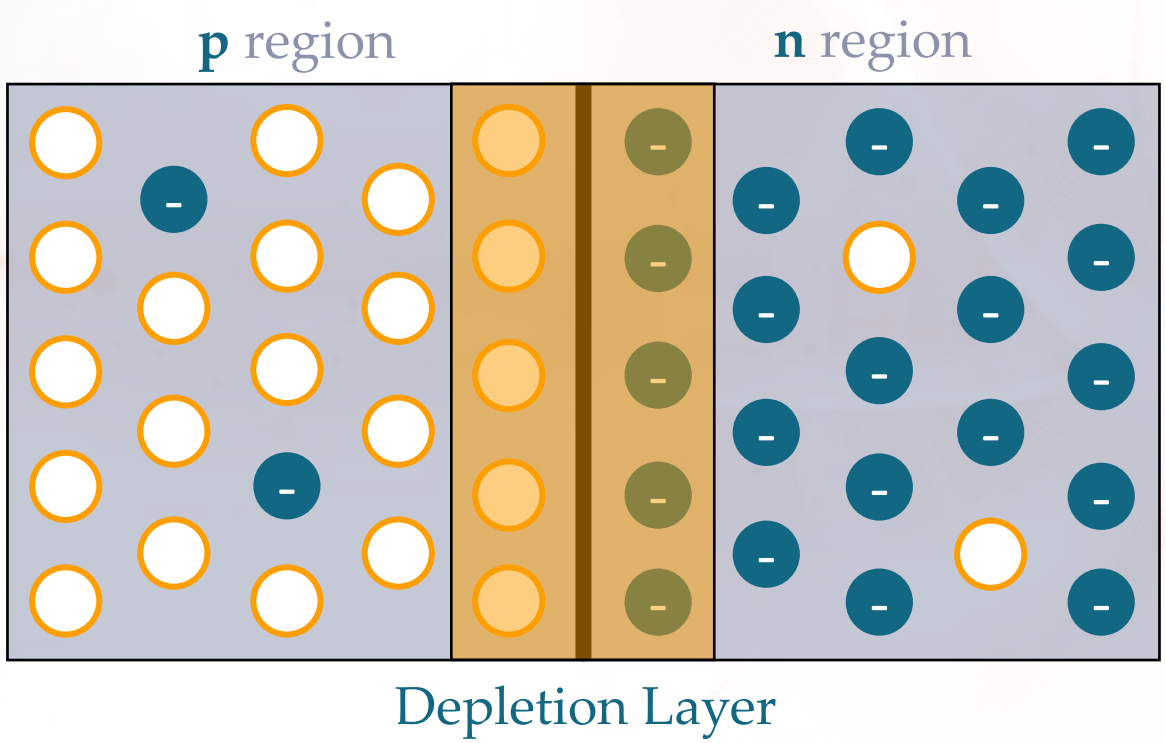
Construction of **two electrode**

DEPLETION LAYER



PN Junction

DEPLETION LAYER



Recombination

The process in which electrons and holes combine

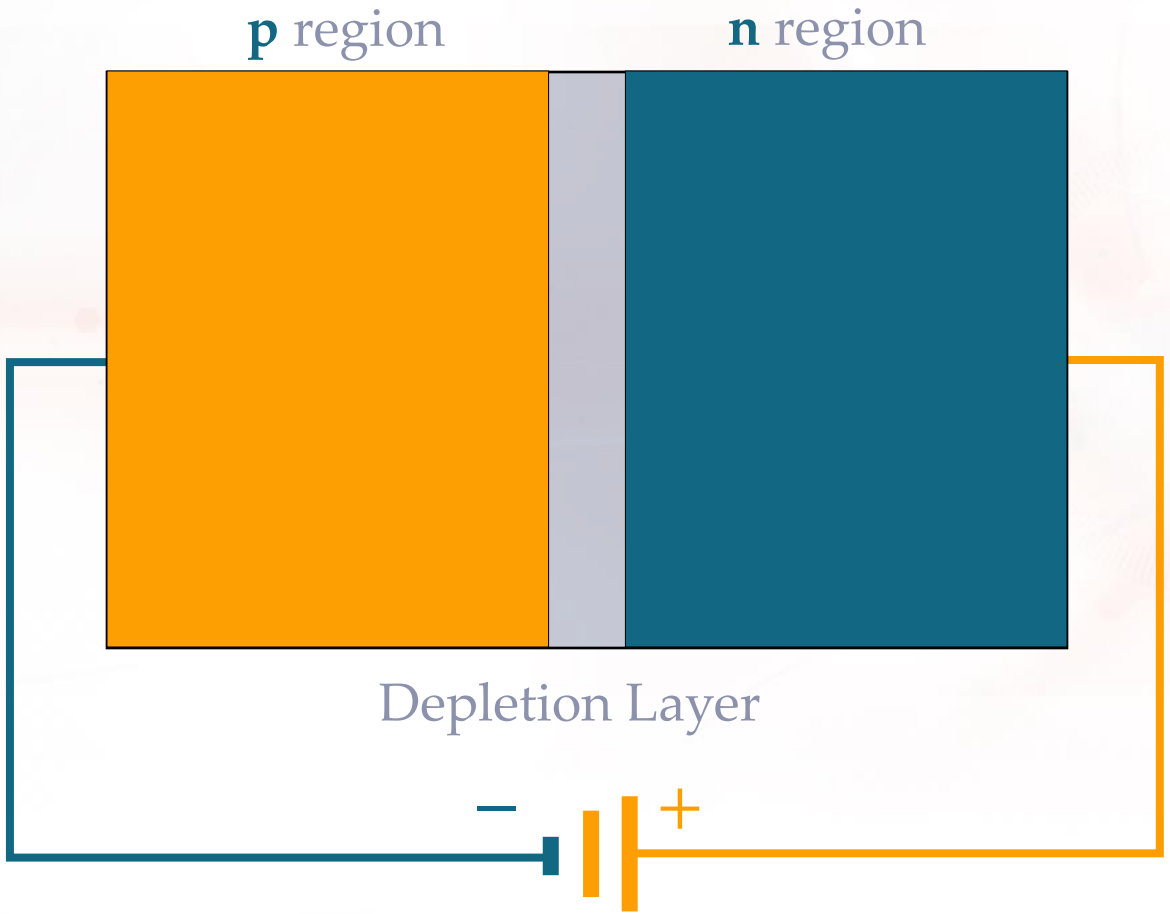
Barrier Potential

The electric field between the ions

0.3V for Germanium (Ge)

0.7V for Silicon (**Si**)

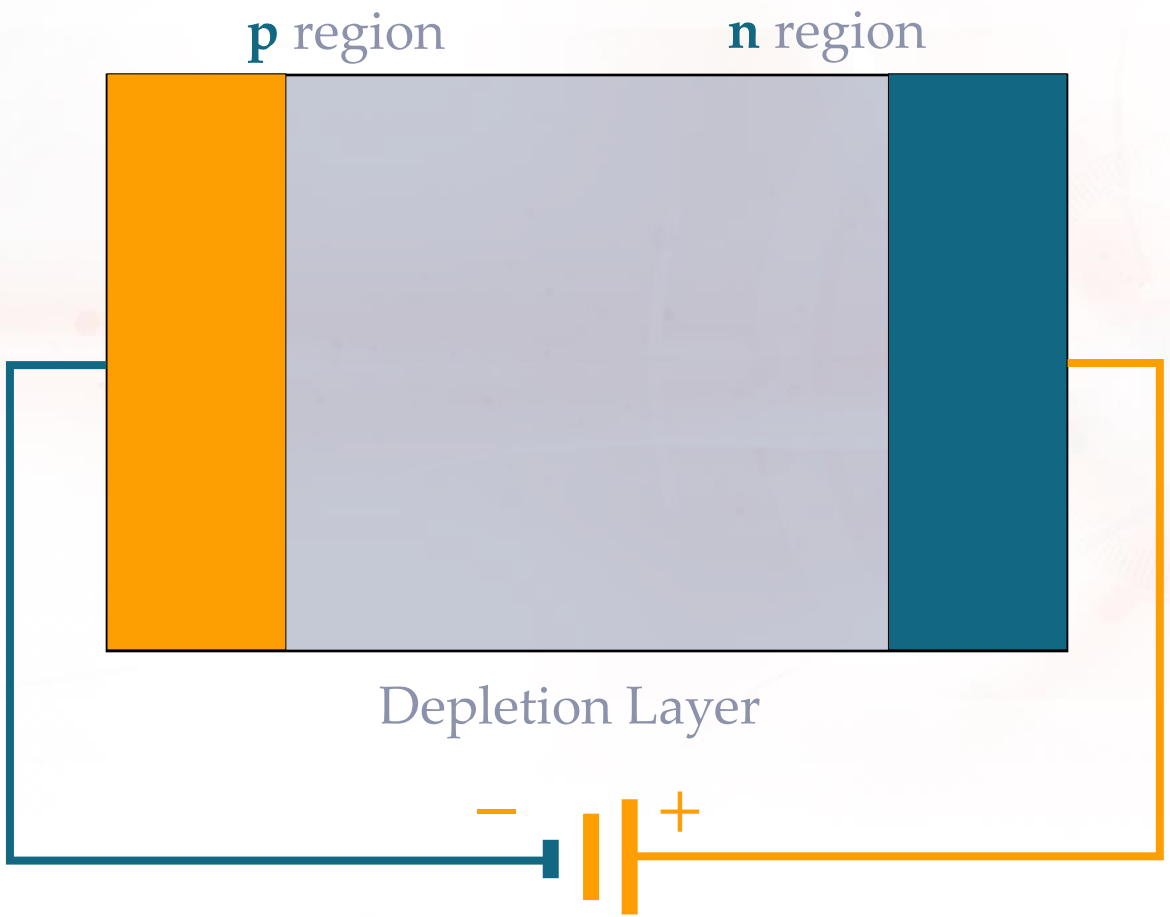
THE BIASED DIODE



Bias

application of **power source**

REVERSE BIAS



Bias

application of **power source**

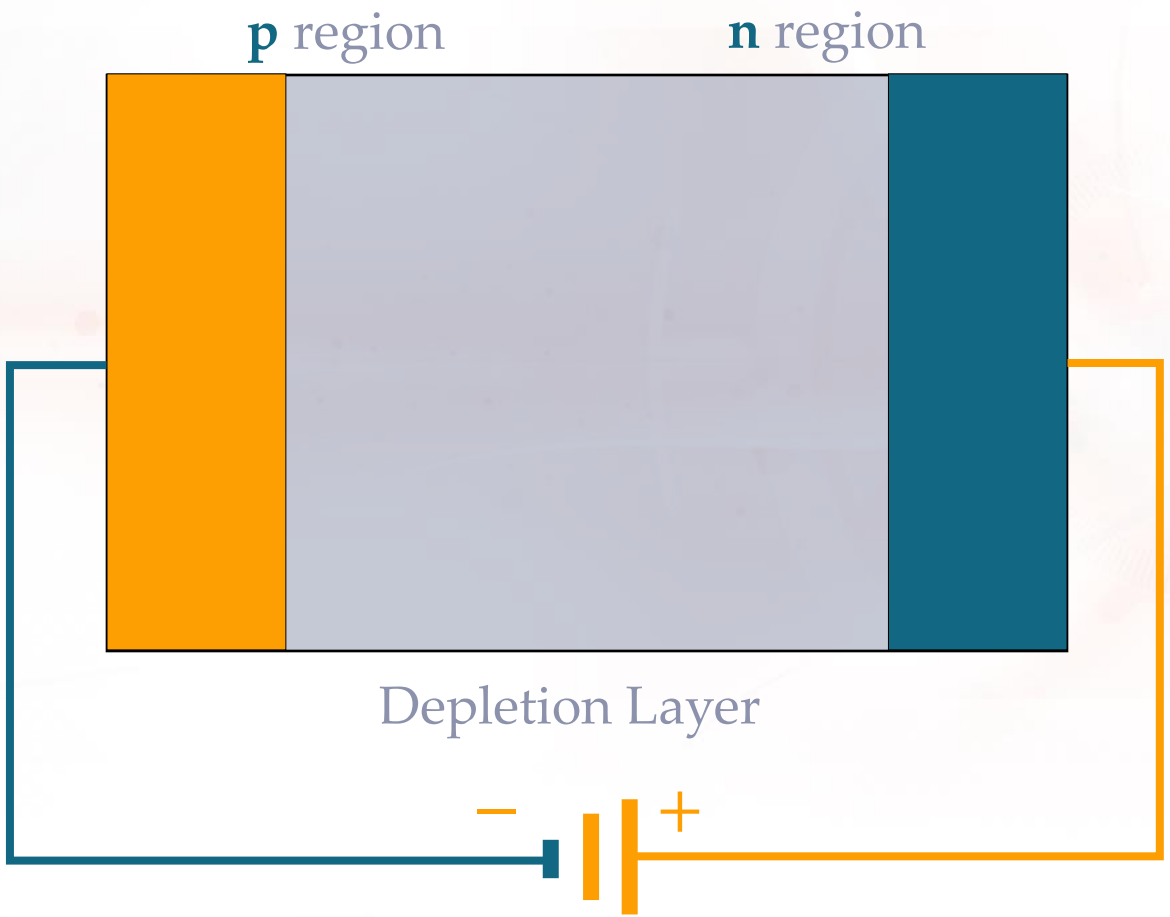
Reversed Saturation Current

The **reversed current** caused by the thermally produced minority carriers

Surface-leakage Current

Caused by the surface **impurities** and **imperfections** in the crystal structure

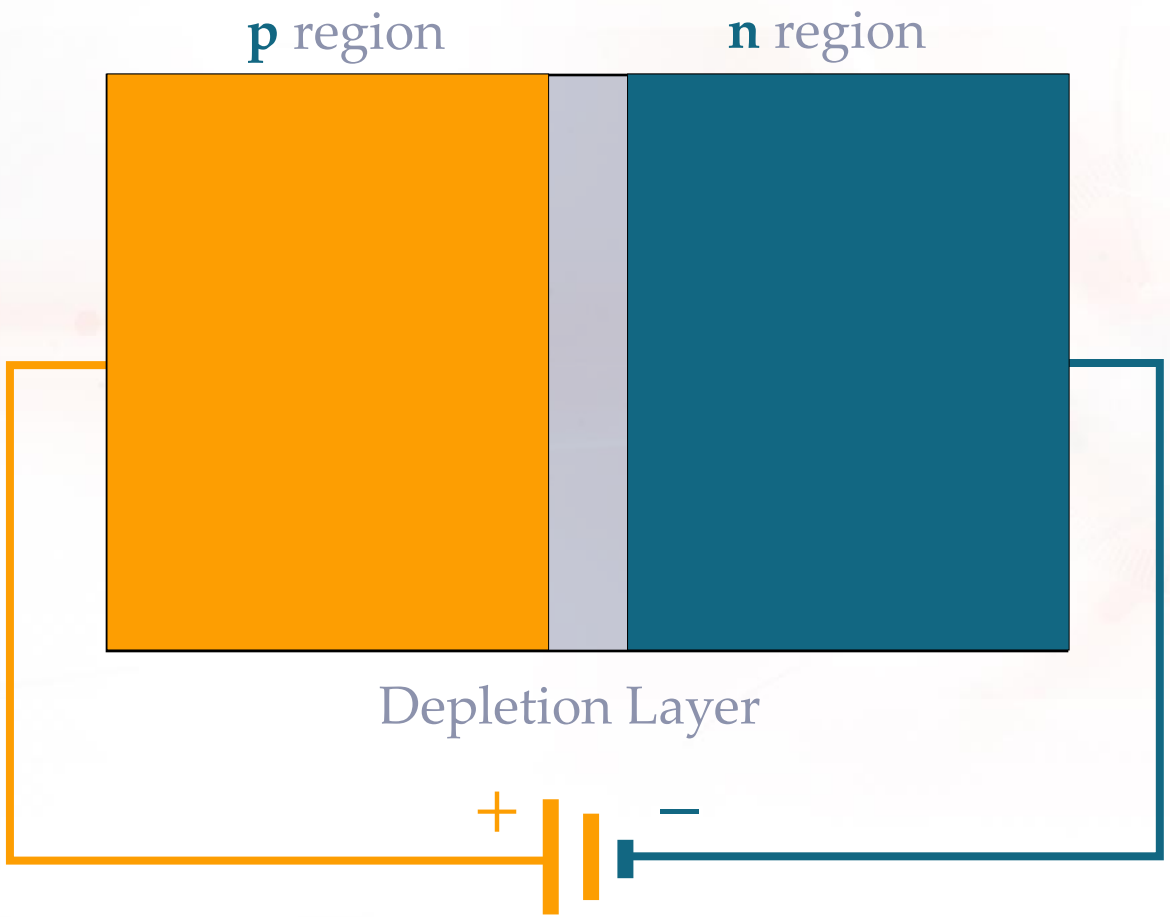
BREAKDOWN VOLTAGE



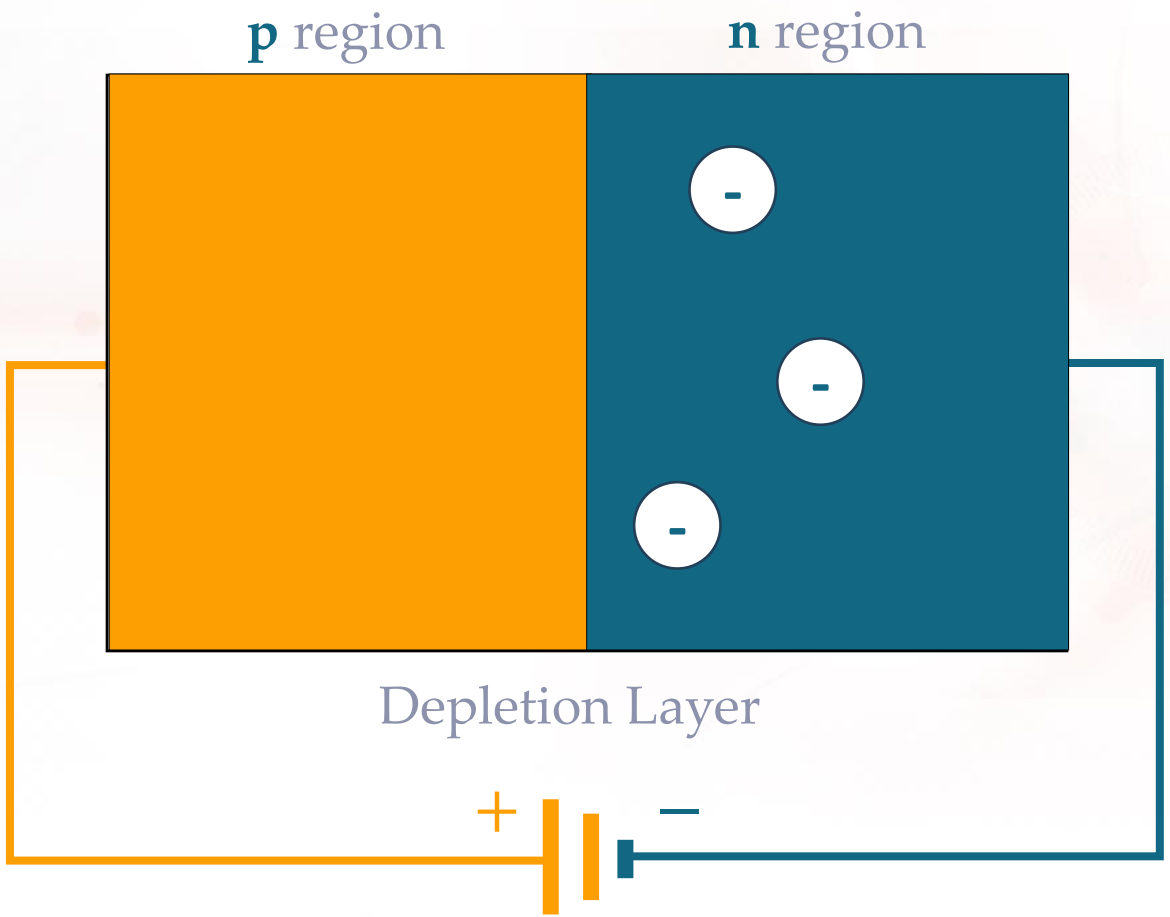
The maximum reverse voltage a diode can withstand before it is **destroyed**

50V (typ)

FORWARD BIAS



FORWARD BIAS

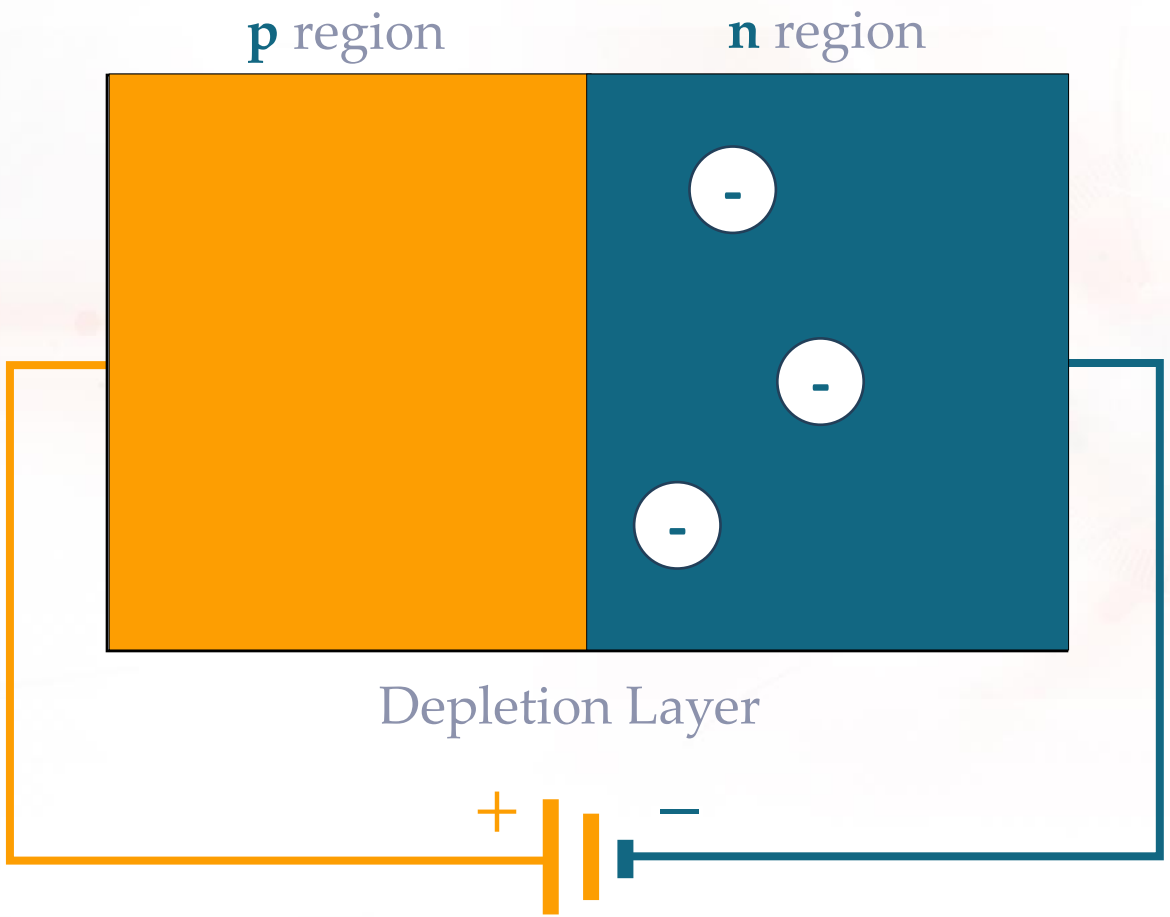


Current flows easily as long as the applied voltage is **greater** than the barrier potential

0.3V (Ge)

0.7V (Si)

FORWARD BIAS



Current flows easily as long as the applied voltage is **greater** than the barrier potential

0.3V (Ge)

0.7V (Si)

DIODE PARAMETERS

TEMPERATURE

Ambient Temperature

air **outside** the diode

Junction Temperature

Inside the diode

@ Conducting

Junction Temp > Ambient Temp

BARRIER POTENTIAL

Depends on the junction temperature

$$\frac{\Delta V}{\Delta T} = -2\text{mV}/°\text{C}$$

EXERCISE

Assuming a barrier potential of **0.7 V** at an ambient temperature of **25°C**. Determine the **barrier potential** of a silicon diode when the junction temperature is

100°C

0°C

50°C

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100°C

0°C

50°C

@100°C,

$$\frac{\Delta V}{\Delta T} = \frac{-2mV}{°C}$$

$$\frac{v_{100°C} - 0.7V}{100°C - 25°C} = \frac{-2mV}{°C}$$

$$\frac{v_{100°C} - 0.7V}{75°C} = \frac{-2mV}{°C}$$

$$v_{100°C} = -2mV(75) + 0.7$$

$$\boxed{v_{100°C} = 0.55\ V}$$

ans

Assuming a barrier potential of **0.7 V** at an ambient temperature of **25°C**. Determine the **barrier potential** of a silicon diode when the junction temperature is

100°C

0°C

50°C

@0°C,

$$\frac{\Delta V}{\Delta T} = \frac{-2mV}{°C}$$

$$\frac{v_{0°C} - 0.7V}{0°C - 25°C} = \frac{-2mV}{°C}$$

$$\frac{v_{0°C} - 0.7V}{-25°C} = \frac{-2mV}{°C}$$

$$v_{0°C} = -2mV(-25) + 0.7$$

$$\boxed{v_{0°C} = 0.75\,V}$$

ans

Assuming a barrier potential of **0.7 V** at an ambient temperature of **25°C**. Determine the **barrier potential** of a silicon diode when the junction temperature is

100°C

0°C

50°C

@50°C,

$$\frac{\Delta V}{\Delta T} = \frac{-2mV}{°C}$$

$$\frac{v_{50°C} - 0.7V}{50°C - 25°C} = \frac{-2mV}{°C}$$

$$\frac{v_{50°C} - 0.7V}{25°C} = \frac{-2mV}{°C}$$

$$v_{50°C} = -2mV(25) + 0.7$$

$$\boxed{v_{50°C} = 0.65\ V}$$

ans

REVERSE SATURATION CURRENT, I_s

I_s **doubles** for each 10°C rise

$$\%\Delta I_s = 100\% \text{ per } 10°C$$

The higher the junction temperature, the greater the saturation current

If the change in temperature are less than 10°C

$$\%\Delta I_s = 7\% \text{ per } °C$$

EXERCISE

A silicon has a saturation current of **5nA** at **25°C**. What is the **saturation current** at **100°C**? At **80°C**?

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@100°C,

25°C	$5\ nA$
35°C	$10\ nA$
45°C	$20\ nA$
55°C	$40\ nA$
65°C	$80\ nA$
75°C	$160\ nA$
85°C	$320\ nA$
95°C	$640\ nA$
96°C	$(640\ nA) + (640\ nA)(0.07)$

doubles

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96°C	$(640\ nA)\ (1.07)$

doubles

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@100°C,

25°C	5 nA	
35°C	10 nA	
45°C	20 nA	
55°C	40 nA	doubles
65°C	80 nA	
75°C	160 nA	
85°C	320 nA	
95°C	640 nA	
96°C	684.8 nA	
97°C	732.7 nA	
98°C	784.0 nA	7%
99°C	838.9 nA	
100°C	$\boxed{897.6\ nA}$	

ans

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@100°C,

25°C	5 nA	
35°C	10 nA	
45°C	20 nA	
55°C	40 nA	2^n
65°C	80 nA	
75°C	160 nA	
85°C	320 nA	
95°C	640 nA	
96°C	684.8 nA	
97°C	732.7 nA	
98°C	784.0 nA	7%
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ans

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85°C	320 nA	
95°C	640 nA	
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97°C	732.7 nA	
98°C	784.0 nA	1.07
99°C	838.9 nA	
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A silicon has a saturation current of **5nA** at **25°C**. What is the **saturation current** at **100°C**? At **80°C**?

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65°C	$80\ nA$	
75°C	$160\ nA$	
85°C	$320\ nA$	
95°C	$640\ nA$	$(5\ nA)2^7$
96°C	$684.8\ nA$	
97°C	$732.7\ nA$	
98°C	$784.0\ nA$	
99°C	$838.9\ nA$	
100°C	$\boxed{897.6\ nA}$	

ans

A silicon has a saturation current of **5nA** at **25°C**. What is the **saturation current** at **100°C**? At **80°C**?

@100°C,

$$\Delta T = 100°C - 25°C$$

$$\Delta T = 75°C$$

$$\Delta T = 7(10°C) + 5(1°C)$$

@95°C,

$$I_s = (5nA)2^7$$

$$I_s = 640\ nA$$

@100°C,

$$I_s = (640nA)1.07^5$$

$$\boxed{I_s = 897.6\ nA}$$

ans

A silicon has a saturation current of **5nA** at **25°C**. What is the **saturation current** at **100°C**? At **80°C**?

@80°C,

$$\Delta T = 80°C - 25°C$$

$$\Delta T = 55°C$$

$$\Delta T = 5(10°C) + 5(1°C)$$

@75°C,

$$I_s = (5nA)2^5$$

$$I_s = 160\ nA$$

@80°C,

$$I_s = (160nA)1.07^5$$

$$\boxed{I_s = 224.4\ nA}$$

ans

SURFACE-LEAKAGE CURRENT, I_{SL}

Directly proportional to the **reverse voltage**

Surface-Leakage Resistance, R_{SL}

$$R_{SL} = \frac{V_R}{I_{SL}}$$

EXERCISE

If the surface leakage current is **2nA** for a reverse voltage of **25V**, what is the **surface-leakage current** for a reverse voltage of **35V**? of **100V**?

$$R_{SL} = \frac{V_R}{I_{SL}}$$

$$R_{SL} = \frac{25V}{2nA}$$

$$R_{SL} = 12.5\ G\Omega$$

If the surface leakage current is **2nA** for a reverse voltage of **25V**, what is the **surface-leakage current** for a reverse voltage of **35V**? of **100V**?

$$R_{SL} = \frac{V_R}{I_{SL}}$$

$$R_{SL} = \frac{25V}{2nA}$$

$$R_{SL} = 12.5\ G\Omega$$

@$35V$,

$$I_{SL} = \frac{V_R}{R_{SL}}$$

$$I_{SL} = \frac{35}{12.5 \times 10^9}$$

$$\boxed{I_{SL} = 2.8\ nA}$$

ans

If the surface leakage current is **2nA** for a reverse voltage of **25V**, what is the **surface-leakage current** for a reverse voltage of **35V**? of **100V**?

$$R_{SL} = \frac{V_R}{I_{SL}}$$

$$R_{SL} = \frac{25V}{2nA}$$

$$R_{SL} = 12.5\ G\Omega$$

$@100V,$

$$I_{SL} = \frac{V_R}{R_{SL}}$$

$$I_{SL} = \frac{100}{12.5 \times 10^9}$$

$$\boxed{I_{SL} = 8\ nA}$$

ans

RECTIFIER CIRCUITS

HALF-WAVE RECTIFIER

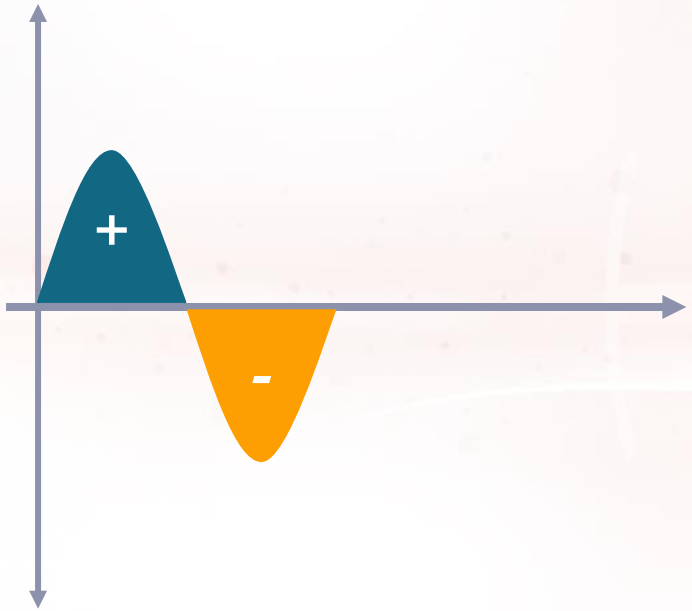
Input

Forward-Biased

Output

HALF-WAVE RECTIFIER

Input

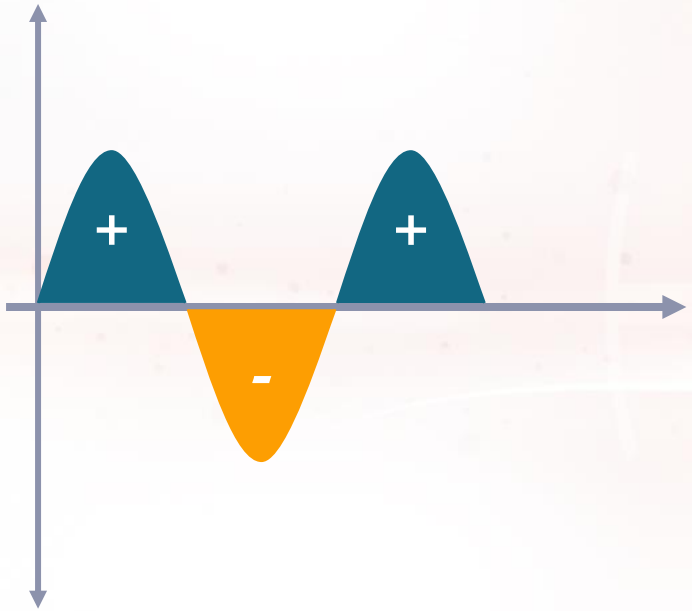


Reverse-Biased

Output

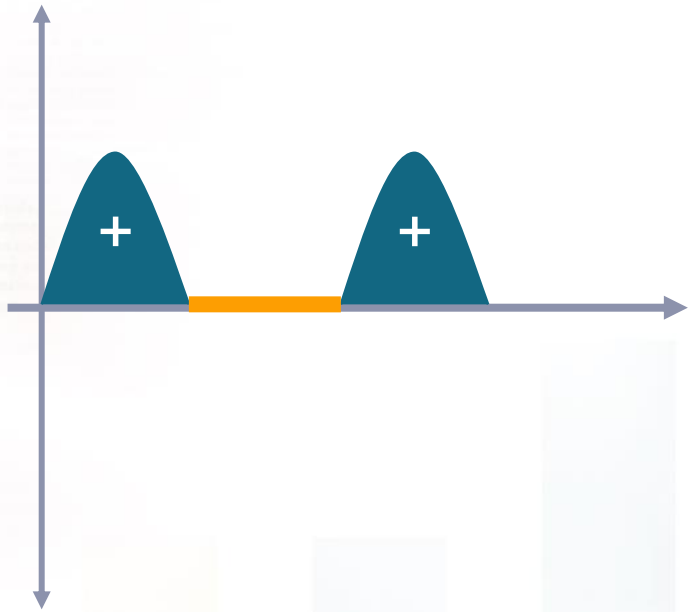
HALF-WAVE RECTIFIER

Input



Forward-Biased

Output



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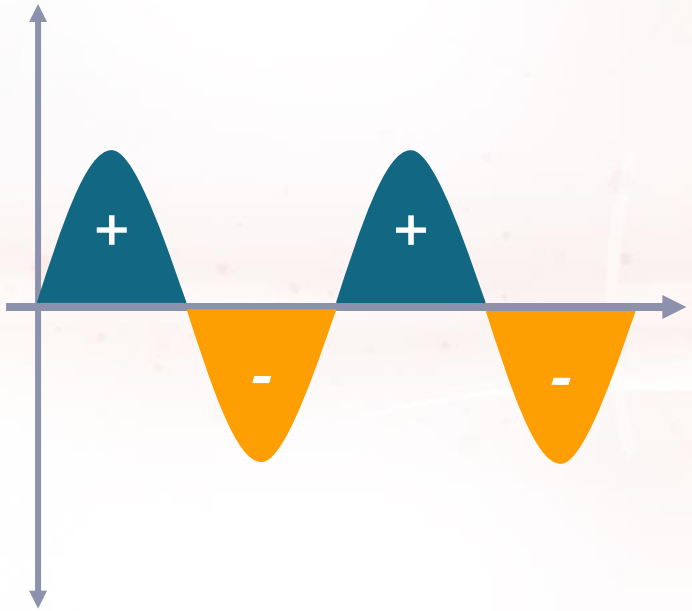
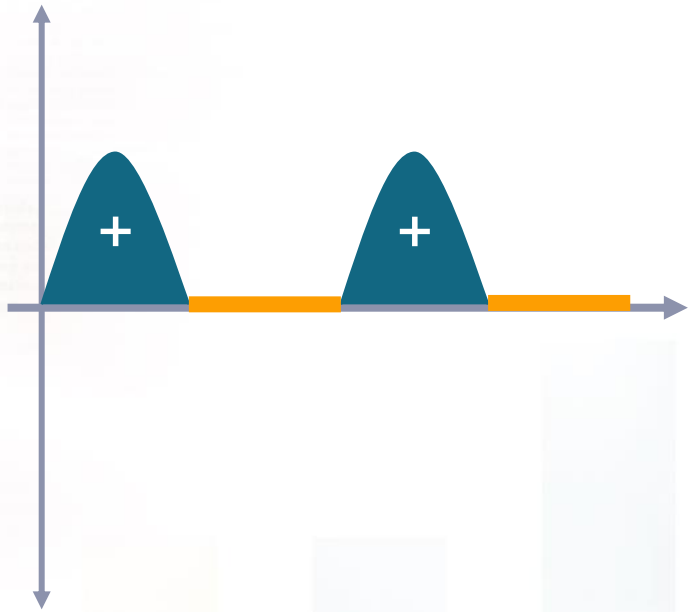


Reverse-Biased

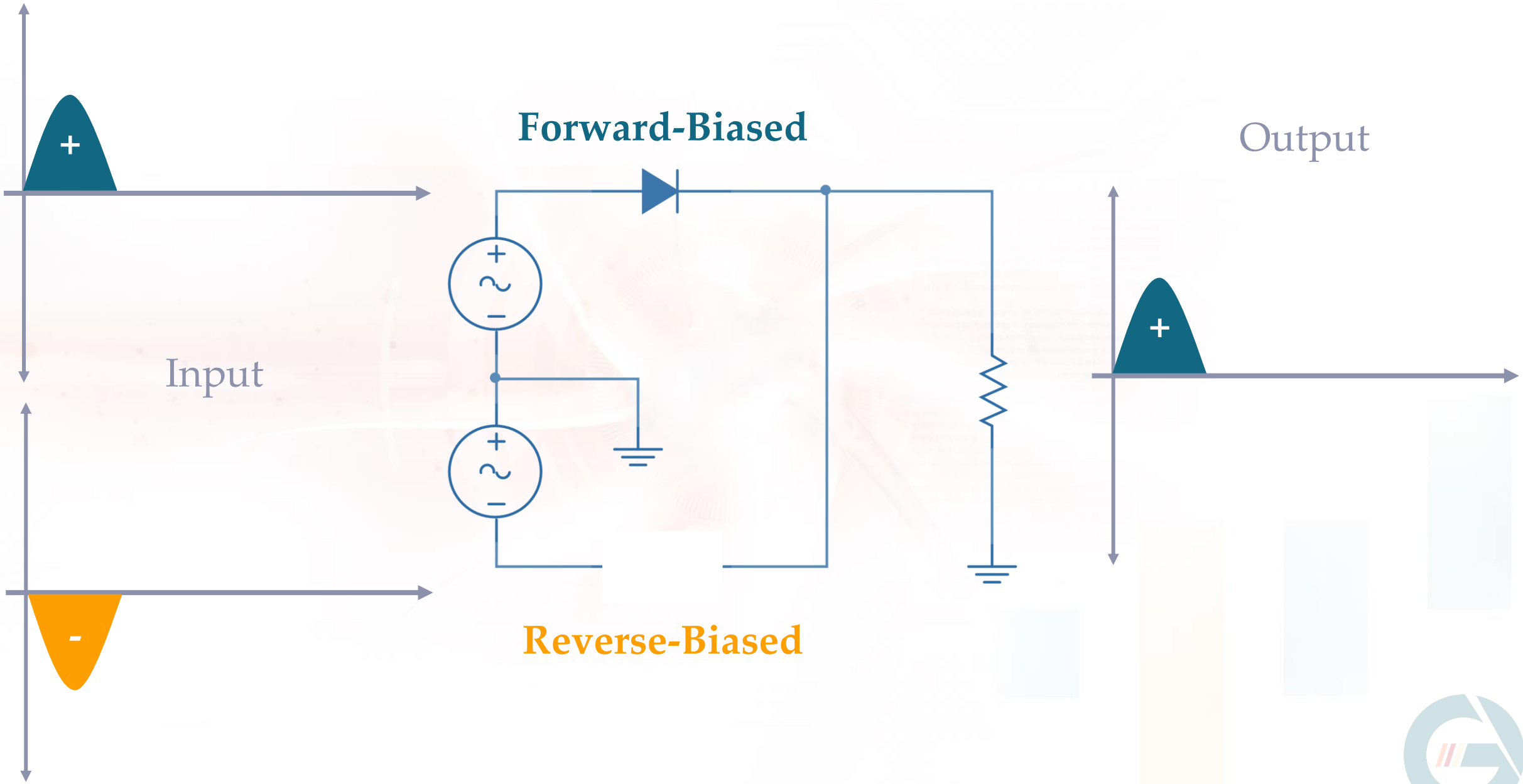
Output



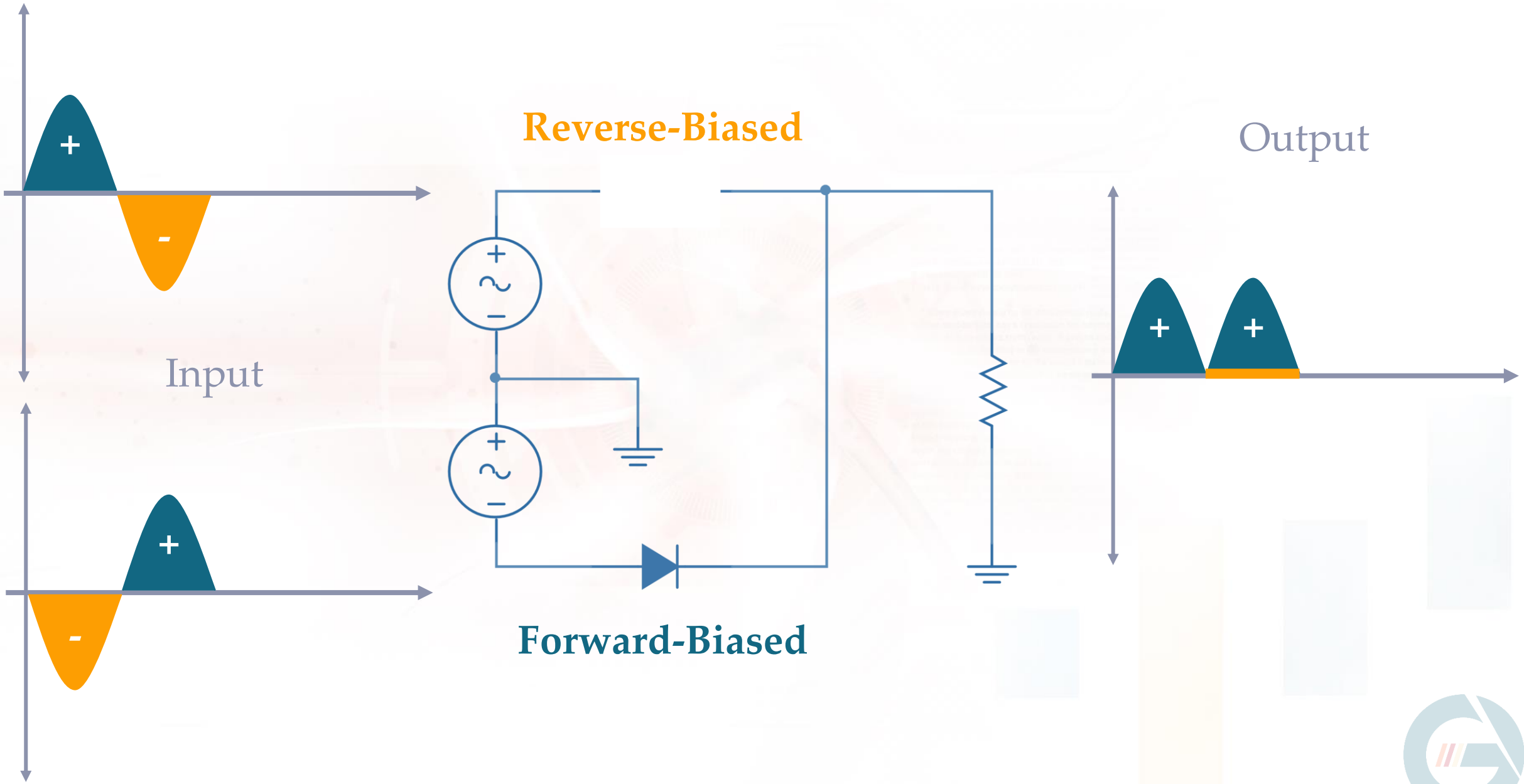
Rectifier

A circuit that converts **AC** voltage to **DC** voltage

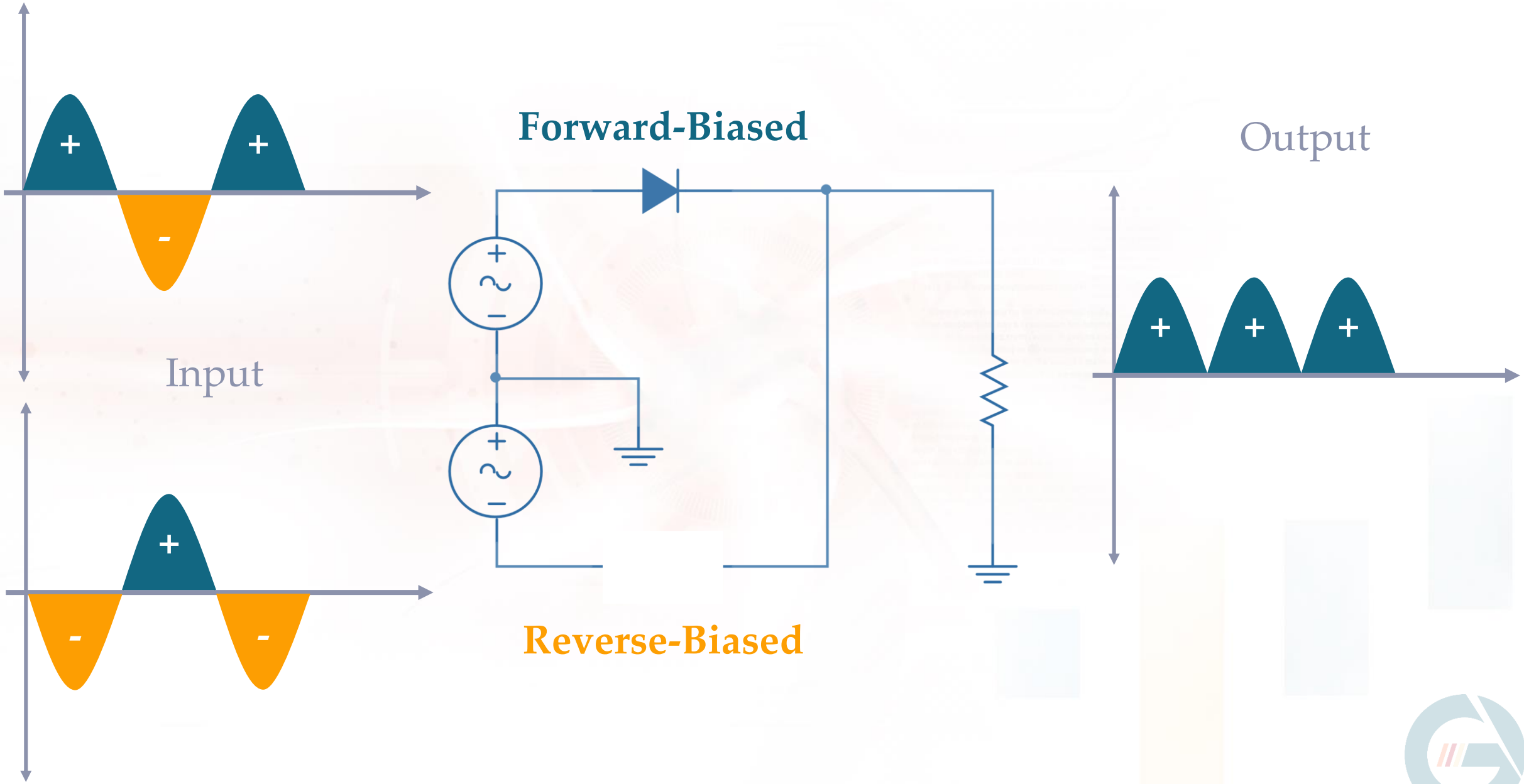
FULL-WAVE CENTER-TAPPED RECTIFIER



FULL-WAVE CENTER-TAPPED RECTIFIER



FULL-WAVE CENTER-TAPPED RECTIFIER



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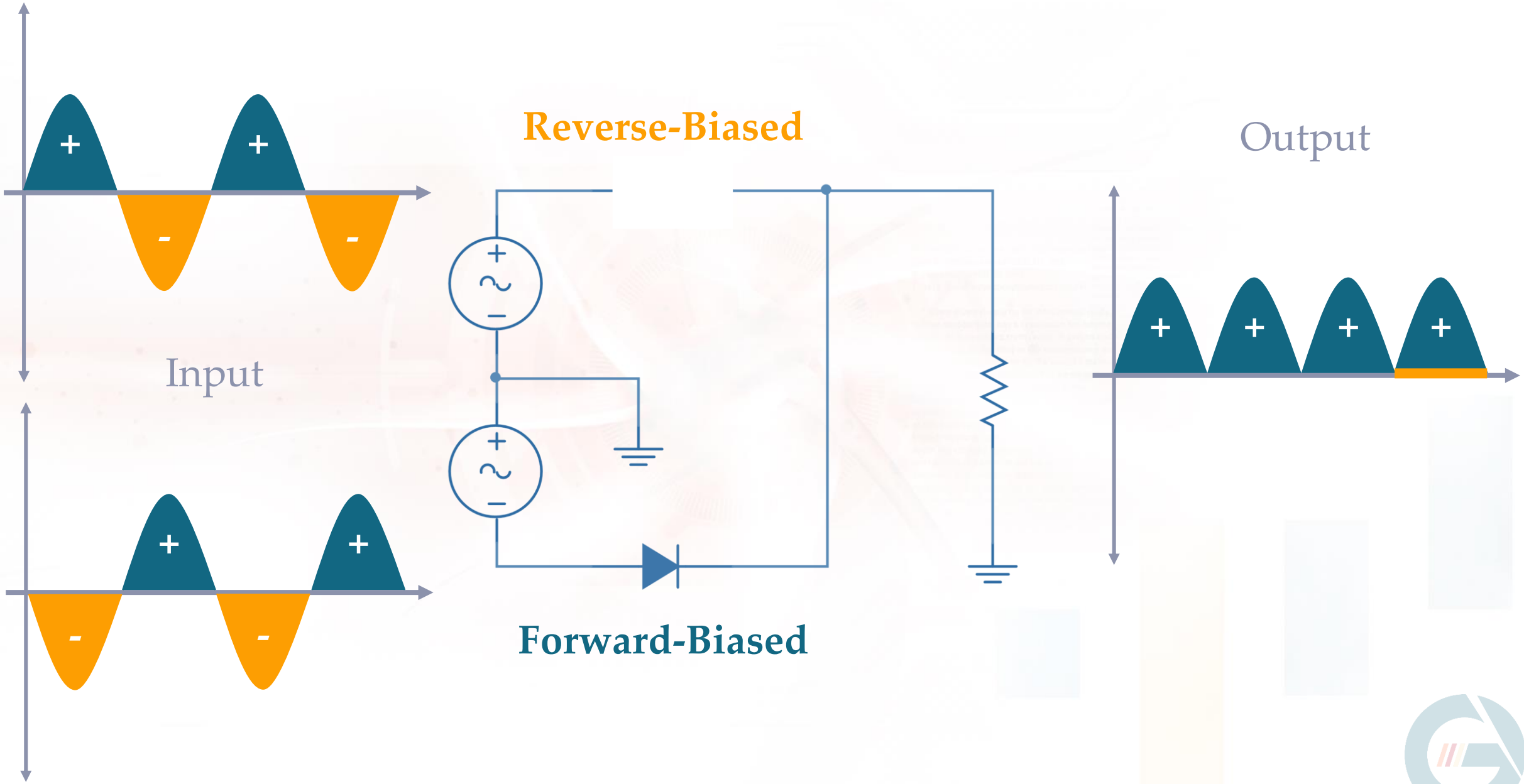
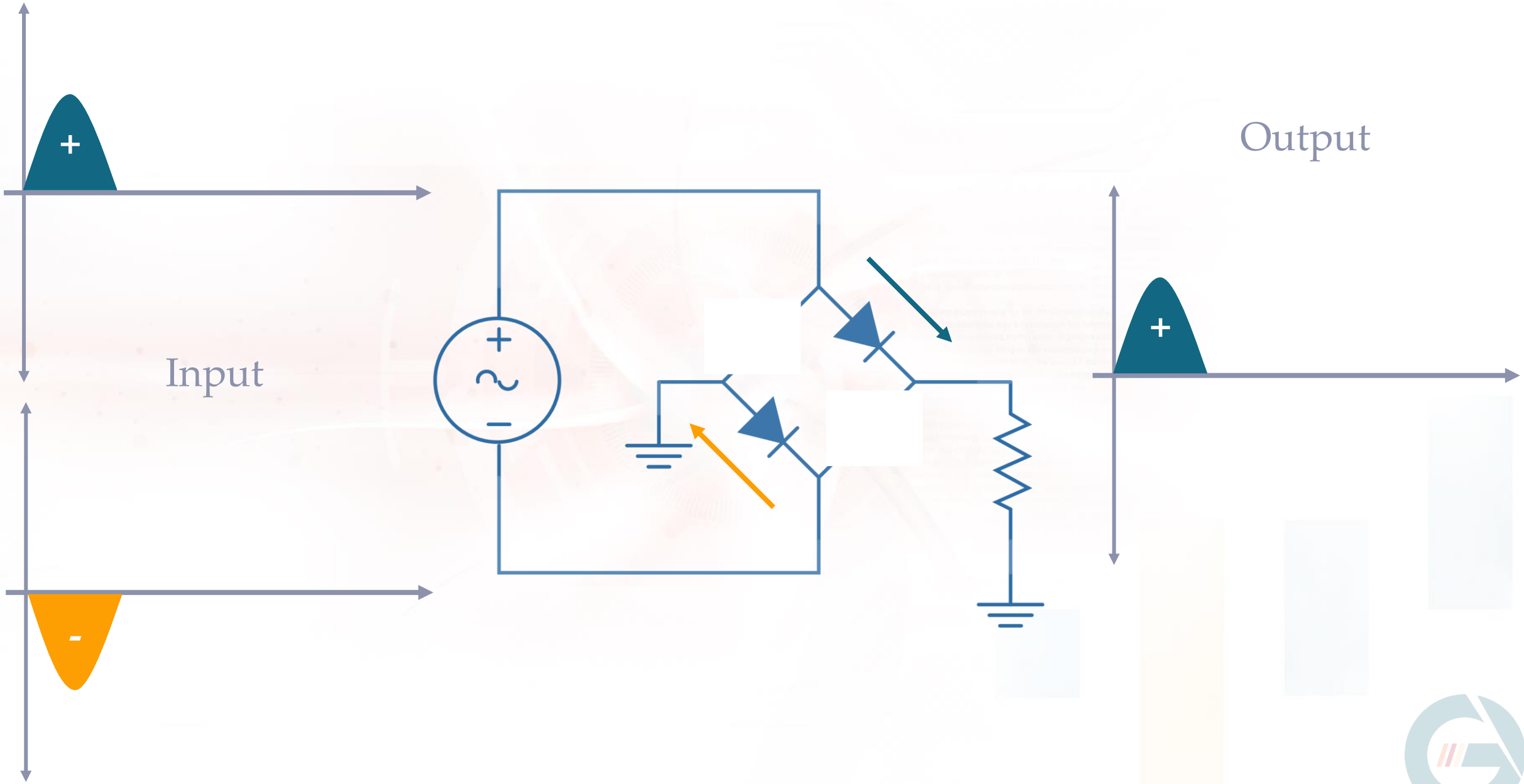


FULL-WAVE BRIDGE TYPE RECTIFIER



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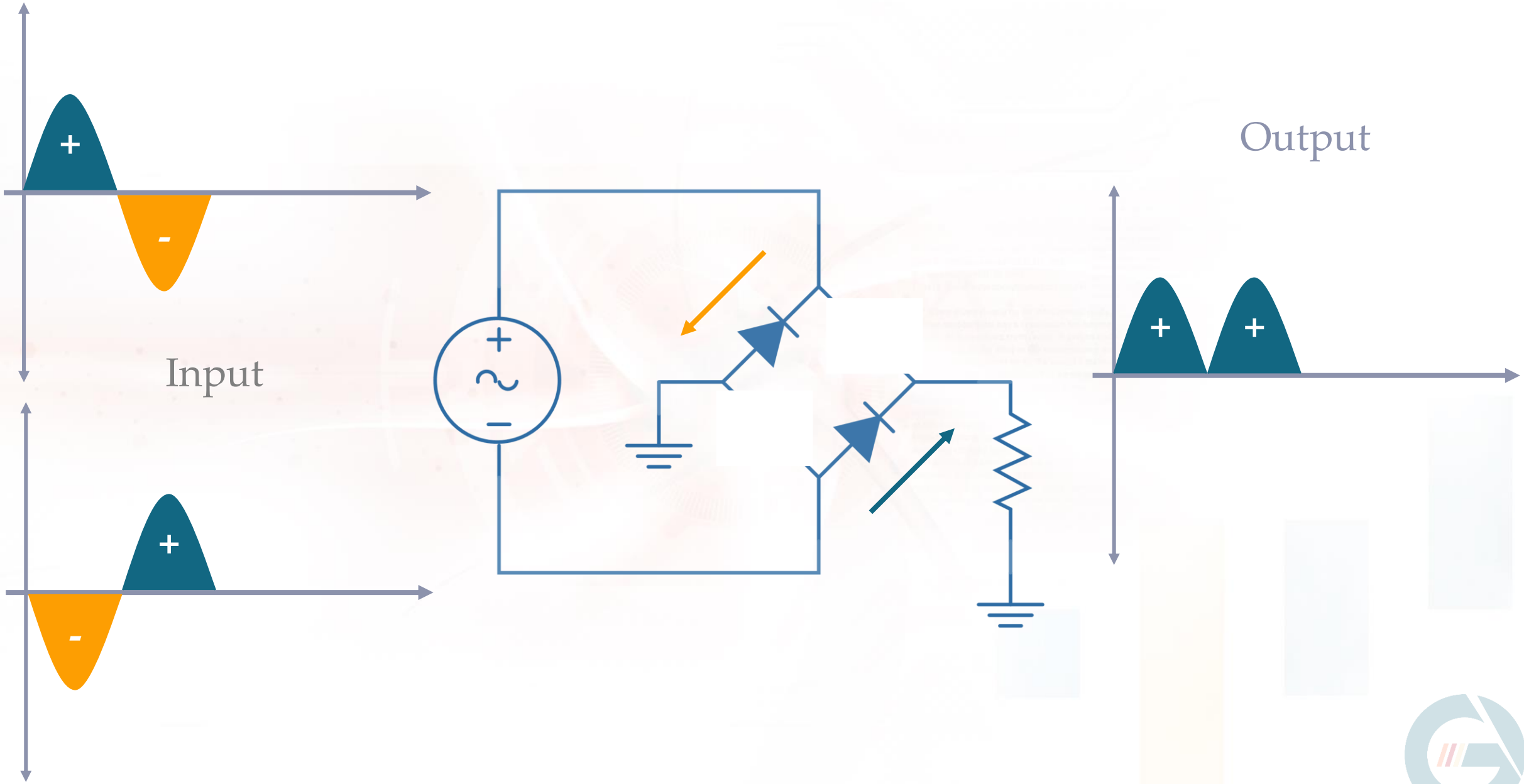

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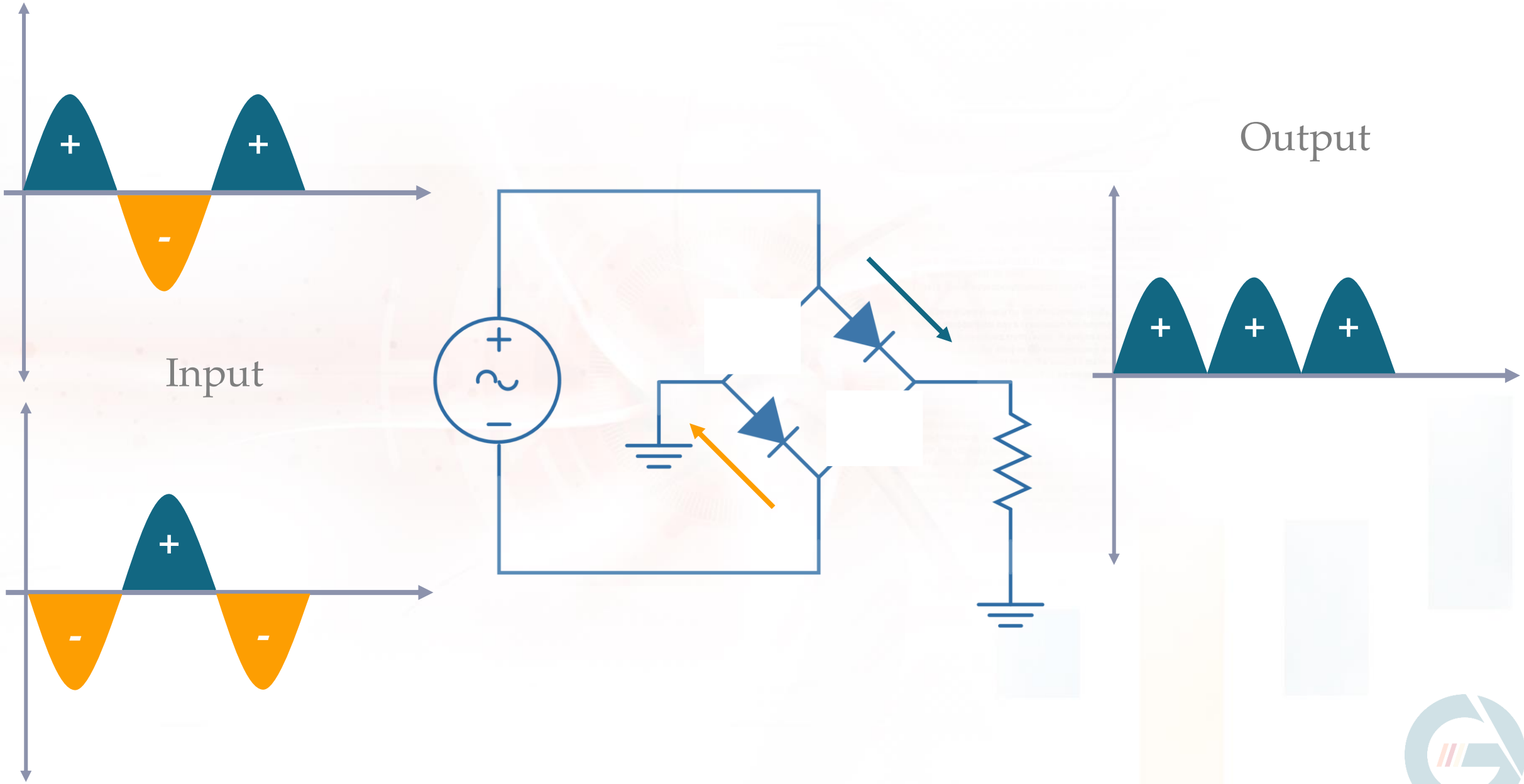
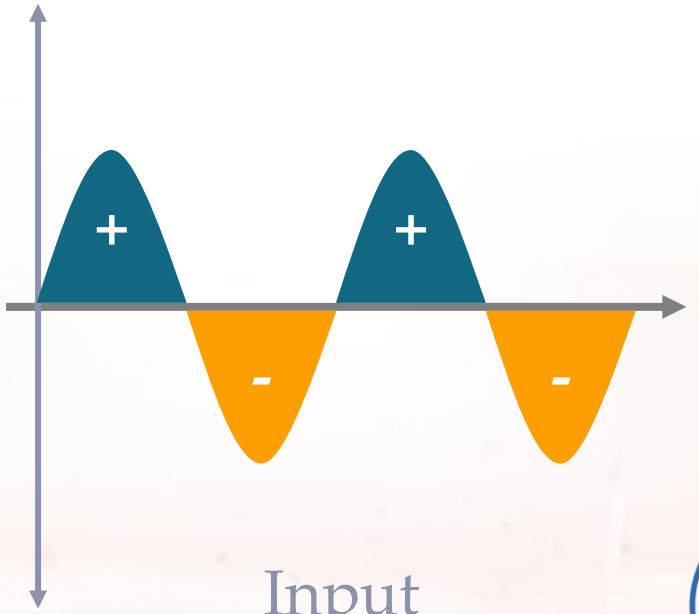
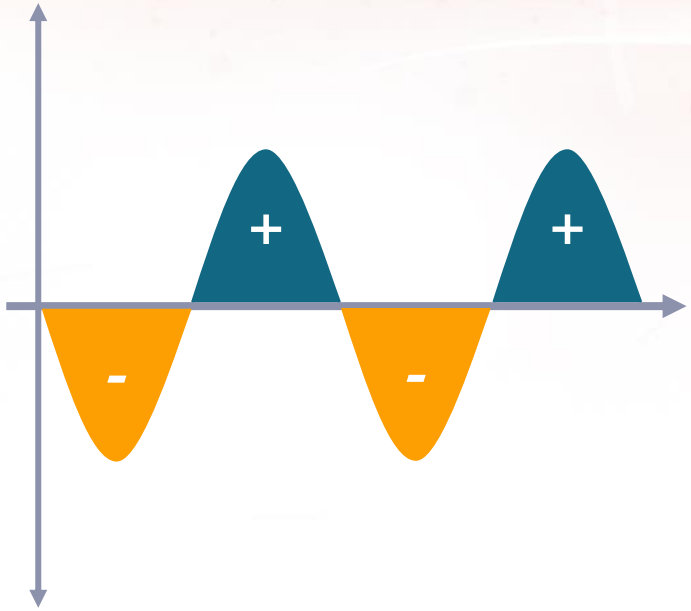
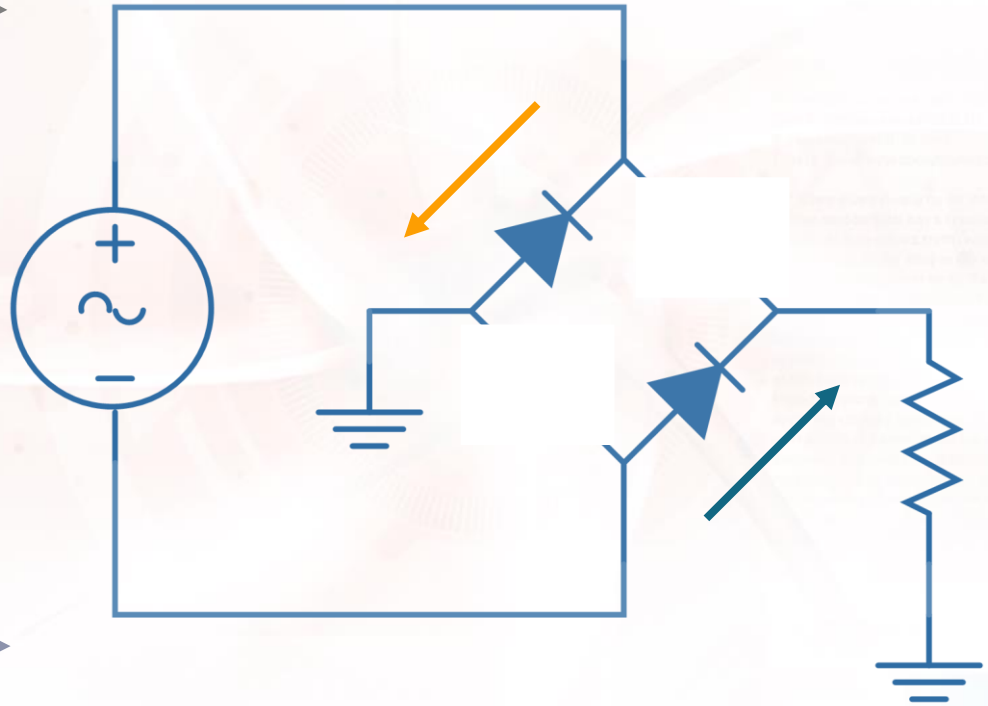


FULL-WAVE BRIDGE TYPE RECTIFIER

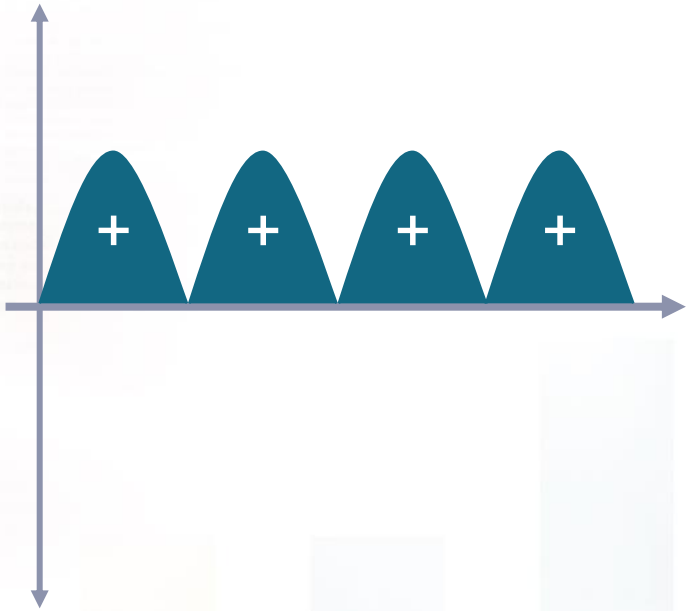


Input





Output



LABORATORY